Outline
Introduction
Objectives
Literature Survey
Implementation using MobileNetV2
Implementation using MobileNetV2
Implementation using Llama2 + RAG
WORKFLOW
Results and Output
Results and Output
Future Work
Conclusion

Multimodal Chatbot for Plant Disease Recognition

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Outline

Outline
Introduction
Objectives
Literature Survey
Implementation using MobileNetV2
Implementation using MobileNetV2
Implementation using Llama2 + RAG
WORKFLOW
Results and Output
Results and Output
Future Work
Conclusion

- Introduction
- Problem Statement
- Objectives
- Literature Survey
- Implementation using MobileNetV2
- Implementation using Llama2 + RAG
- Work Flow
- Results and Outputs
- Future Work
- Conclusion
- Bibliography



Introduction

Outline
Introduction
Objectives
Literature Survey
Implementation using MobileNetV2
Implementation using MobileNetV2
Implementation using Llama2 + RAG
WORKFLOW
Results and Output
Results and Output
Future Work
Conclusion

- Agriculture is a critical sector for global food security. Plant diseases can significantly impact crop yield and quality.
- Farmers often struggle to identify diseases promptly due to a lack of access to expert knowledge.
- A chatbot can bridge this gap by providing instant disease recognition and treatment suggestions.



Objectives

Outline
Introduction
Objectives
Literature Survey
Implementation using MobileNetV2
Implementation using MobileNetV2
Implementation using Llama2 + RAG
WORKFLOW
Results and Output
Results and Output
Future Work
Conclusion

The major objectives of the project are as follows:-

- Identify plant diseases from images of leaves.
- Provide text-based diagnosis and treatment suggestions.
- Retrieve relevant information from a knowledge base using Retrieval-Augmented Generation (RAG).
- Ensure fast and accurate responses with a FAISS-based vector search engine.



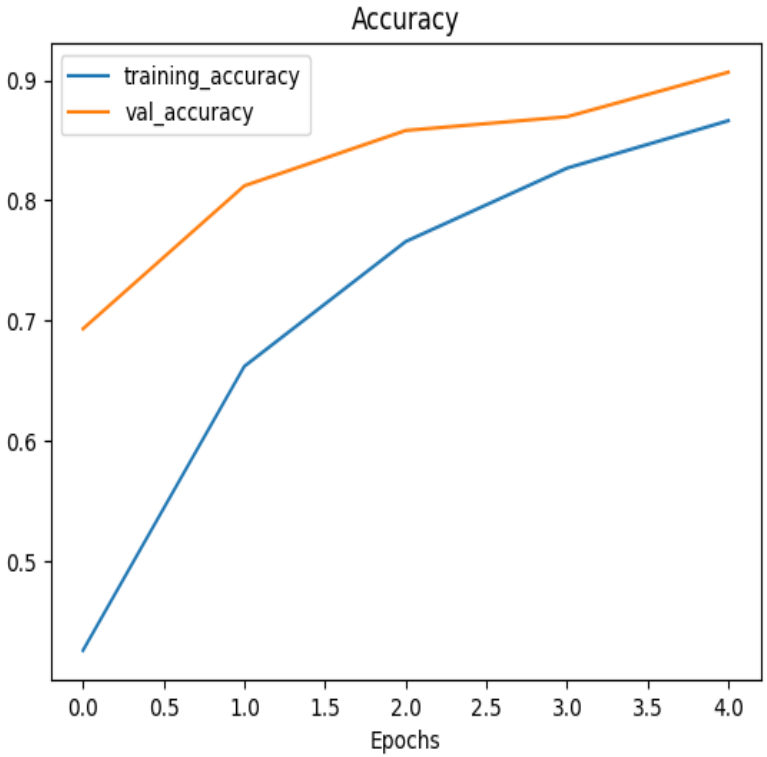
Literature Survey

Outline
Introduction
Objectives
Literature Survey
Implementation using MobileNetV2
Implementation using MobileNetV2
Implementation using Llama2 + RAG
WORKFLOW
Results and Output
Results and Output
Future Work
Conclusion

It reviews various studies focused on applying machine learning and deep learning techniques for plant disease detection, highlighting the datasets, methods, and challenges involved. Many of the studies utilize commonly available datasets such as the PlantVillage dataset[1], along with specialized datasets like the Indian Medicinal Leaves and Potato Leaf Disease datasets.[2] The methods employed range from Convolutional Neural Networks (CNNs) and Transfer Learning to advanced architectures like AELGNet, MDSCIRNet, and CoffeeNet. These models often achieve high accuracy rates, with some surpassing 90 percentage[3], showcasing their potential for real-time disease detection.

Implementation using MobileNetV2

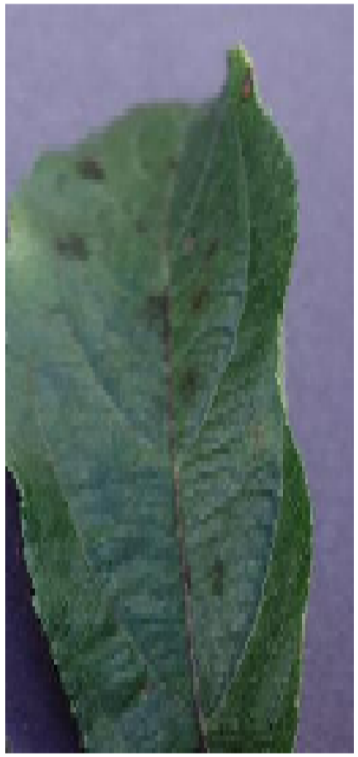
Outline
Introduction
Objectives
Literature Survey
Implementation using MobileNetV2
Implementation using MobileNetV2
Implementation using Llama2 + RAG
WORKFLOW
Results and Output
Results and Output
Future Work
Conclusion

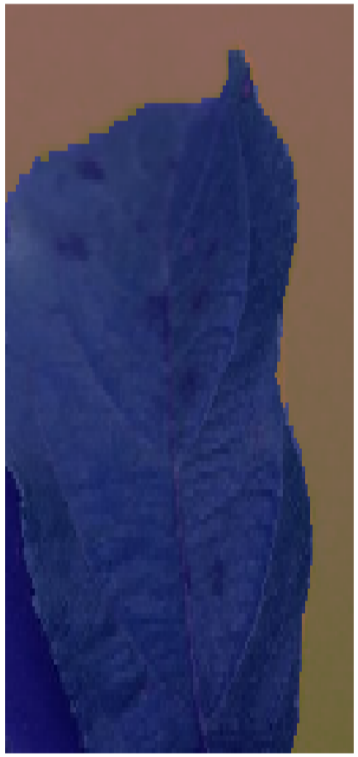




Implementation using MobileNetV2

Outline
Introduction
Objectives
Literature Survey
Implementation using MobileNetV2
Implementation using MobileNetV2
Implementation using Llama2 + RAG
WORKFLOW
Results and Output
Results and Output
Future Work
Conclusion





Metric	Value	Remarks
Training Accuracy	98.2%	Indicates strong learning capability on the dataset.
Validation Accuracy	95.1%	Suggests good generalization on unseen data, with minor overfitting.
F1-Score	¿0.90 for most classes	High precision and recall across multiple disease classes.
Training Loss	0.12	Low loss demonstrates effective optimization during training.
Validation Loss	0.22	Slightly higher than training loss, showing generalization with some room for improvement.



Implementation using Llama2 + RAG

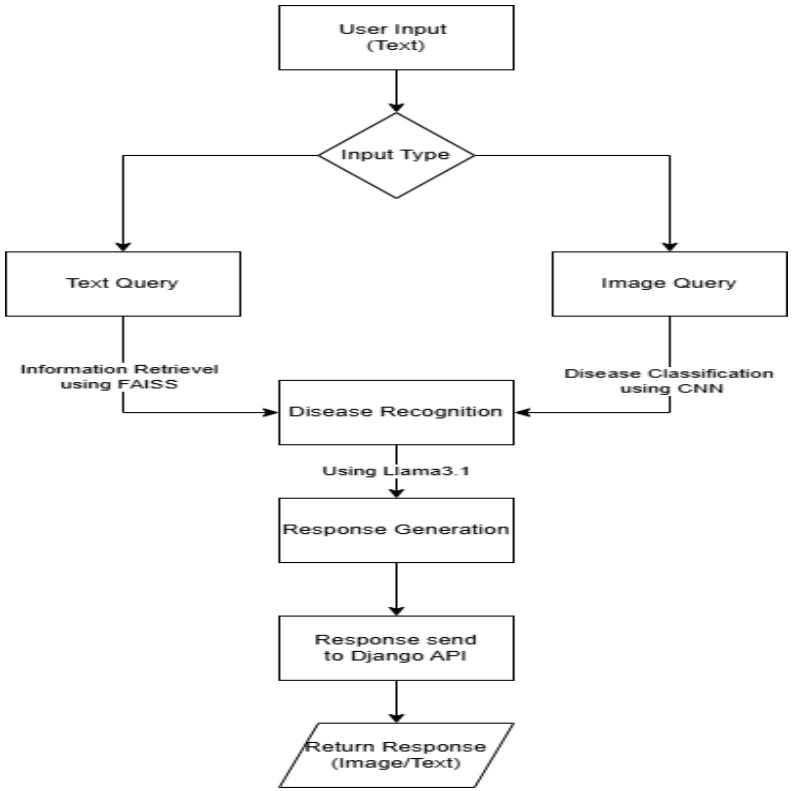
Outline
Introduction
Objectives
Literature Survey
Implementation using MobileNetV2
Implementation using MobileNetV2
Implementation using Llama2 + RAG
WORKFLOW
Results and Output
Results and Output
Future Work
Conclusion

Components of the system:-

- Text-based disease retrieval: FAISS-powered semantic search for disease-related information.
- Image-based disease recognition: Deep learning model (CNN) classifies plant diseases.
- LLaMA 2 chatbot: Generates detailed responses based on retrieved knowledge.
- Django API: Handles user interaction and integration.

WORKFLOW

Outline
Introduction
Objectives
Literature Survey
Implementation using MobileNetV2
Implementation using MobileNetV2
Implementation using Llama2 + RAG
WORKFLOW
Results and Output
Results and Output
Future Work
Conclusion



Results and Outputs

Outline

Introduction

Objectives

Literature Survey

Implementation using MobileNetV2

Implementation using MobileNetV2

Implementation using Llama2 + RAG

WORKFLOW

Results and Output

Results and Output

Future Work

Conclusion





Results and Outputs

Outline
Introduction
Objectives
Literature Survey
Implementation using MobileNetV2
Implementation using MobileNetV2
Implementation using Llama2 + RAG
WORKFLOW
Results and Output
Results and Output
Future Work
Conclusion


```
Welcome to the Plant Disease Chatbot. Type 'exit' to quit.
User: What are common leaf diseases in corn?
Bot: Based on the following plant disease information: 5, 5, 5, 5, 5. Answer the question: What are common leaf diseases in cor

5 - Leaf blight
5 - Rust
5 - Powdery mildew
5 - Fungal infections
5 - Root rot

The common leaf diseases in corn are:

* Leaf blight
* Rust
* Powdery mildew

Note: The answer is based on the information provided in the question.
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Future Work

Outline
Introduction
Objectives
Literature Survey
Implementation using MobileNetV2
Implementation using MobileNetV2
Implementation using Llama2 + RAG
WORKFLOW
Results and Output
Results and Output
Future Work
Conclusion

Components of the system:-

- Voice-based and Image-based chatbot interaction.
- Real-time database updates from agricultural research.
- Mobile app development for farmers.
- Deployment of project on cloud.



Conclusion

Outline
Introduction
Objectives
Literature Survey
Implementation using MobileNetV2
Implementation using MobileNetV2
Implementation using Llama2 + RAG
WORKFLOW
Results and Output
Results and Output
Future Work
Conclusion

- The multimodal chatbot for plant disease recognition combines FAISS-based text retrieval, CNN-based image classification, and LLaMA 2 for generating intelligent responses. Integrated with a Django API, it provides an efficient and user-friendly system for diagnosing plant diseases. This solution can help farmers and researchers improve crop health and yield. Future improvements may include support for more plant species, multilingual support, and real-time applications.



REFERENCES

Outline

Introduction

Objectives

Literature Survey

Implementation using MobileNetV2

Implementation using MobileNetV2

Implementation using Llama2 + RAG

WORKFLOW

Results and Output

Results and Output

Future Work

Conclusion

M. V. Shewale and R. D. Daruwala, "High performance deep learning architecture for early detection and classification of plant leaf disease," *Journal of Agriculture and Food Research*, vol. 14, p. 100675, 2023.

K. P. Ferentinos, "Computers and electronics in agriculture,"

Z. Zhang, W. Zhan, Y. Sun, J. Peng, Y. Zhang, Y. Guo, K. Sun, and L. Gui, "Mask-guided dual-perception generative adversarial network for synthesizing complex maize diseased leaves to augment datasets," *Engineering Applications of Artificial Intelligence*.